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Design Bending Loads for a Multiple Longeron Structure

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Introduction

The use of a structural concept consisting of longerons and bulkheads to carry bending loads is fairly common in small sounding rockets. The purpose of this memo is to document how the loads are distributed among the longerons.

Another view of essentially the same problem is that of a joint. The joint under consideration here connects two tubes, one sliding inside the other. Their attachment uses n radial fasteners, rivets, bolts, etc., located symmetrically around the tubes. Any bending moment at the joint is carried by fastener shear. The other option, fastener compression/tension, is usually not preferred because many fasteners are stronger in shear than in tension. As a corollary, the two concentric tubes are usually clamped together to some degree near a fastener so that friction can also play a role in carrying bending moment. In fact, when n is large friction is the dominant mechanism and the joint is as stiff as the parent tube. Failure occurs by tearing of the parent tubes between fasteners.

In general, this is a statically indeterminate problem. That is, there are more fastener loads to be estimated than there are static equilibrium constraint equations. We therefore assume that plane cross-sections remain plane after bending.

Nomenclature

Mnemonic	Definition
L	Number of longerons,
A	Cross section area of a longeron,
P_i	Axial load acting on the ith longeron,
E	Young's modulus of the longeron material,
F	Axial force acting on the structure,
ε_i	Axial strain in the ith longeron,
M	Bending moment acting on the structure
R	Radius of a circle passing through the longeron area centroids,
r	Radius of bending curvature,
ϕ_i	The roll angle of the ith longeron,
d_i	Distance from the neutral axis to the areal centroid of the ith longeron, and
j	$\sqrt{-1}$

Analysis

Consider a structure of L identical, equally spaced longerons. Then the ith longeron lies a distance d_i from the neutral axis:

$$d_i = R\sin\phi_i . \tag{1}$$

Neglect the intrinsic bending stiffness of the longerons, and assume the entire bending moment is carried by axial load in the longerons. The axial load action of the ith longeron is given by Hooke's law:

$$P_i = AE\varepsilon_i . \tag{2}$$

Since plane sections are assumed to remain plane after bending, the strain of the ith longeron is $\frac{R\sin\phi_i}{r}$. Thus,

$$P_i = AER\sin\phi_i / r . \tag{3}$$

The ith longeron's contribution to the bending moment is given by

$$M_i = AER^2\sin^2\phi_i / r . \tag{4}$$

Next, we show that there is no net axial force:

$$P = \frac{AER}{r}\sum_1^L \sin\phi_i . \tag{5}$$

Use the complex definition of the sine function:

$$\sin\phi_i = \exp\, j(\phi_1 + \frac{2\pi(i-1)}{L}) = \exp(j\phi_1)\exp(\frac{2\pi j(i-1)}{L}) .$$

(6)

It follows that the phase angle of the first longeron simply acts as a multiplier. The second term above starts off with a longeron on the neutral axis for which the sine vanishes. Each successive longeron is paired with one on the opposite side of the neutral axis so that the two sines cancel each other. If L is even the final longeron is also on the neutral axis.

The situation is different in the case of the bending moment. Here we must assess $\sum_1^L \sin^2\phi_i = \frac{L}{2}$ using an inductive proof . The total bending moment is

$$M = \frac{AER^2L}{2r}, \text{ or}$$

$$M = \frac{AER^2L}{2}\frac{d^2y}{dx^2}. \tag{7}$$

Note that this is equivalent to the bending deflection of a solid section whose cross section moment of inertia I is

$$I = \frac{1}{2}AR^2L$$

Finally, the maximum axial load acting on a longeron is

$$P_{MAX} = \pm\frac{2M}{RL}. \tag{8}$$

Equation (8) is the preferred result to be used in a stress analysis of a longeron-bulkhead structure. The $\pm$ sign indicates that some longerons are in compression while others are in tension.

If the structure is required to carry an axial force due to, say, acceleration, then these loads must be added to the bending moment load. If all longerons carry axial force equally,

$$P = F/L \tag{9}$$

Then the appropriate combination of eq's. (8) and (9) should be used in stress analysis.

Monocoque Shell Joints

Consider two circular monocoque shells joined by rivets, bolts, etc. acting in shear. The design shear loads carried by the most heavily loaded fastener are given by eq'a. (8) and (9) if there are L fasteners and P is the design shear force.

That's all, folks!